§2.10 Warm-Up: Taylor & Maclaurin Series

Name: _____ Date: _____

Maclaurin ($a = 0$): $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$ Taylor about a: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$

1. Maclaurin series for $f(x) = \frac{1}{2-x}$.

n	$f^{(n)}(x)$	$f^{(n)}(0)$	$\frac{f^{(n)}(0)}{n!}$
0			
1			
2			
3			

$T_3(x) =$ _____

2. Maclaurin series for $f(x) = \frac{1}{1+x^2}$.

n	$f^{(n)}(x)$	$f^{(n)}(0)$	$\frac{f^{(n)}(0)}{n!}$
0			
1			
2			
3			

$T_3(x) =$ _____

3. Taylor series for $f(x) = \sqrt{x}$, centered at $a = 4$. (center $a \neq 0$)

n	$f^{(n)}(x)$	$f^{(n)}(4)$	$\dfrac{f^{(n)}(4)}{n!}$
0			
1			
2			
3			

$T_3(x) =$ _____

(Remember: powers of $(x-4)$, not powers of x.)

Answer Key **(teacher — do not print for students)**

1. $f(x) = (2-x)^{-1}$: $f' = (2-x)^{-2},\ f'' = 2(2-x)^{-3},\ f''' = 6(2-x)^{-4}$;
so $f(0) = \frac{1}{2},\ f'(0) = \frac{1}{4},\ f''(0) = \frac{1}{4},\ f'''(0) = \frac{3}{8}$.

$$T_3(x) = \frac{1}{2} + \frac{1}{4}x + \frac{1}{8}x^2 + \frac{1}{16}x^3 \qquad \left(\text{general term } \sum_{n=0}^{\infty} \frac{x^n}{2^{n+1}}\right)$$

2. $f(x) = (1+x^2)^{-1}$: $f' = -2x(1+x^2)^{-2}$, then product/chain rule gives $f''(0) = -2$;
f is even, so all odd derivatives vanish: $f(0) = 1,\ f'(0) = 0,\ f''(0) = -2,\ f'''(0) = 0$.

$$T_3(x) = 1 - x^2 \qquad \left(\text{general term } \sum_{n=0}^{\infty} (-1)^n x^{2n} = 1 - x^2 + x^4 - \cdots\right)$$

3. $f(x) = \sqrt{x}$ at $a = 4$: $f(4) = 2,\ f'(4) = \frac{1}{4},\ f''(4) = -\frac{1}{32},\ f'''(4) = \frac{3}{256}$.

$$T_3(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2 + \frac{1}{512}(x-4)^3$$